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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-23

Programme: B. Tech / MBA Tech (AI & Data Science/ AIML/ AI/CSE Cyber Security/Computer/IT/EXTC) Year: II /III Semester: III/V
Subject: Discrete Mathematics

Date: 18 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 04

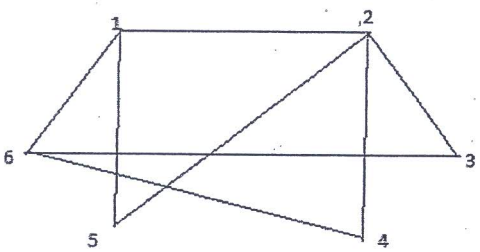
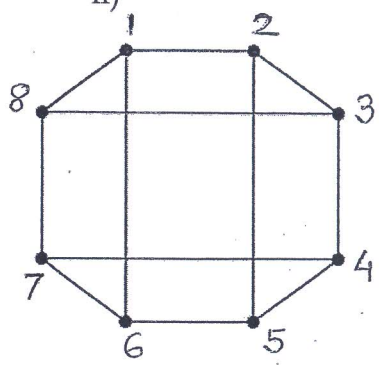
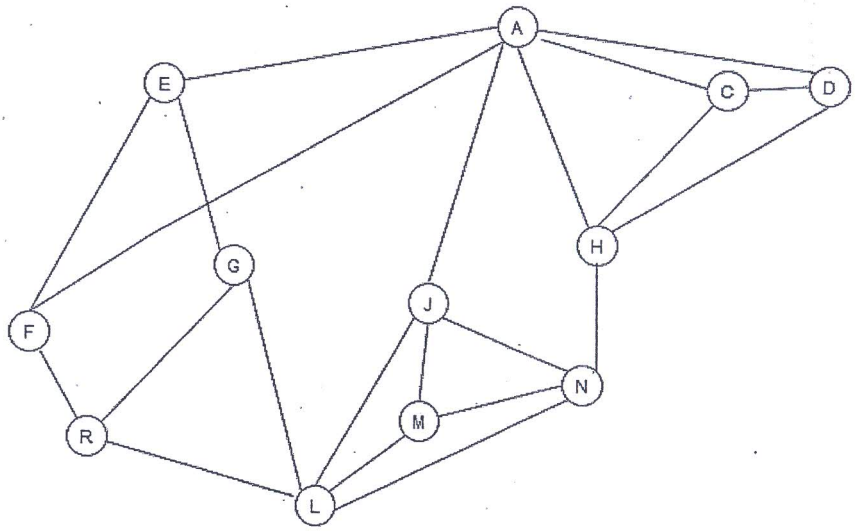
Final Examination

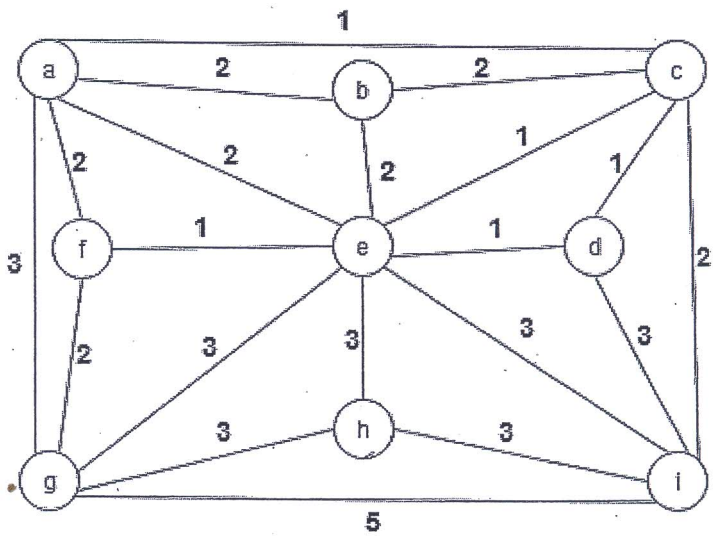
Marks: 100

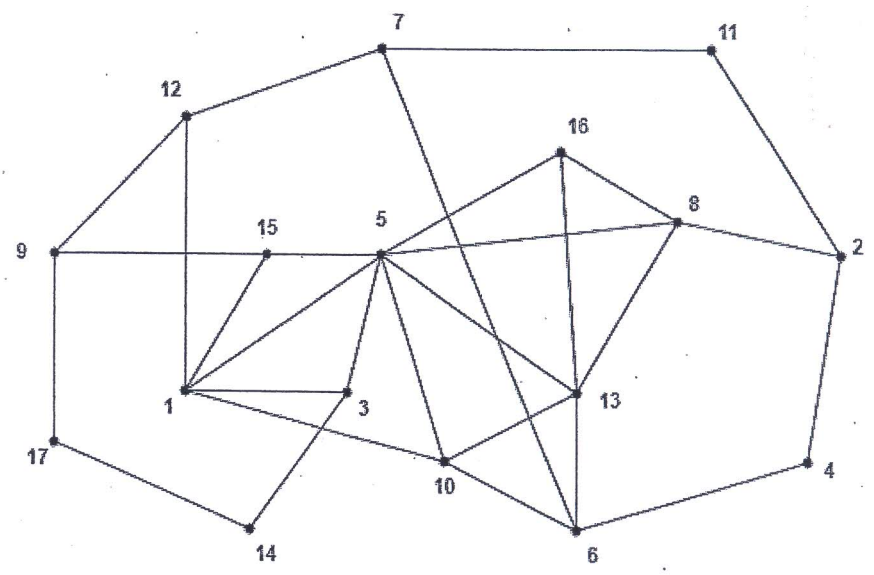
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1. Questions 1 is compulsory.
2. Solve any four out of remaining six questions
3. **In all, 5 questions to be attempted**
4. All questions carry equal marks.
5. **Answer to each new question to be started on a fresh page.**
6. **Figures in brackets on the right hand side indicate full marks.**
7. **Assume Suitable data if necessary.**
8. **Use of scientific calculator is allowed.**

		Answer the following:	Marks
CO1/SO2/BL3	Q.1 a.	If A and B are subsets of the universal set U and $n(U) = 50$ $n(A) = 35,\ n(B) = 20,$ and $n(A' \cap B') = 5$, find i) $n(A \cup B)$ ii) $n(A \cap B)$ iii) $n(A' \cap B)$ iv) $n(A \cap B')$.	[4]
CO2/SO1/BL2	b.	Construct a truth table for $(P \rightarrow Q) \wedge (Q \rightarrow R)$.	[4]
CO1/SO1/BL1	c.	State the generalized pigeon hole principle. Find the minimum number of teachers in a college to be sure that four of them are born in same month.	[4]
CO4/SO1/BL3	d.	Check whether the following graphs exists. Justify your answer. i) A tree with 6 vertices and 5 edges. ii) A graph with degree sequence (2,2,3,3,4)	[4]

CO4/SO2/BL5	e.	Show that $\{\mathbb{R}^+, *\}$ where $*$ is defined by $a*b=\frac{ab}{2}$ for all $a,b\in\mathbb{R}^+$ is a monoid where $\mathbb{R}^+$ is the set of all positive integers.	[4]
CO3/SO2/BL5	Q.2 a.	Solve the recurrence relation: $a_n = 6a_{n-1} - 9a_{n-2}$, $n \geq 3$, $a_1 = -9, a_2 = -9$	[6]
CO1/SO2/BL3	b.	Determine which of the following graphs are bi-partite. Justify your answer. i)  ii)  	[6]
CO3/SO2/BL4	c.	Let $\mathbb{Z}^*$ be a set of non-zero integers and let R be a relation on $A = \mathbb{Z}^* \times \mathbb{Z}^*$ defined by $(a,b)R(c,d)$ *if* $ad = bc$, Then prove that R is an equivalence relation.	[8]
CO2/SO1/BL2	Q.3 a.	Using the principle of mathematical induction show that for all positive integers $n \geq 1$, $5^n - 4n - 1$ is divisible by 16.	[6]
CO2/SO2/BL6	b.	Estimate the chromatic number of the following graph using Welch-Powell Algorithm  	[6]

CO1/SO1/BL4	c.	Let f, g be functions from set of real number to set of real number such that $f(x) = 3x - 6$ and $g(x) = \frac{x}{3} + 3$. Find $f \circ g$ and $g \circ f$. Examine whether $f \circ g$ is bijective or not. If $f \circ g$ is bijective find inverse of $f \circ g$.	[8]
CO3/SO2/BL4	Q.4 a.	Prove that the set of rational numbers $(\mathbb{Q}, \oplus, \otimes)$ is a ring under the operations defined by $a \oplus b = a + b + 7 \quad \& \quad a \otimes b = a + b + \frac{ab}{7}$	[6]
CO2/SO2/BL2	b.	Convert $\sim(p \to q) \vee ((r \wedge s) \to (q \vee t)) \vee (\sim p \to \sim v)$ into Conjunctive Normal Form.	[6]
CO3/SO1/BL5	c.	Solve the recurrence relation : $a_n + 4a_{n-1} + 4a_{n-2} = n + 1$	[8]
CO2/SO1/BL5	Q.5 a.	If G is a group of real numbers under addition and G' is the group of positive real numbers under multiplication, show that the mapping defined by $f(x) = 3^x$ is an isomorphism.	[6]
CO2/SO2/BL4	b.	Prove that $\sqrt{2}$ is an irrational number using method of contradiction.	[6]
CO3/SO1/BL3	c.	Construct a minimum spanning tree using Kruskal's Algorithm. Also find the minimum total weight. 	[8]

CO1/SO2/BL4	Q.6 a.	Let R be a relation on set of real numbers ($\mathbb{R}$), such that aRb if and only if $a^2 + b^2 = 4$. Determine whether the relation R on the set $\mathbb{R}$ is (1) irreflexive (2) antisymmetric (3) transitive with proper justification.	[6]
CO4/SO1/BL3	b.	Identify whether the following graphs are planar or not. Give reason in detail. i) $K_{3,2}$ ii) $K_{3,3}$	[6]
CO4/SO1/BL6	c.	For $\mathbb{Z}_7 - \{0\}$ i) Prepare composition table with respect to multiplication modulo 7 ii) Prove that it is an abelian group with respect to multiplication modulo 7 iii) Is it cyclic? iv) Find the order of 2 & 4 and subgroups generated by these elements.	[8]
CO2/SO2/BL3	Q.7 a.	Apply laws of logic to establish the equivalence: $p \to ((p \to q) \wedge \sim(\sim q \vee \sim p)) \equiv \sim p \vee (q \wedge p)$	[6]
CO3/SO7/BL2	b.	Solve the following recurrence relation $a_n + 4a_{n-1} + 4a_{n-2} = 8$ from n ≥ 3 with $a_1 = 1$, $a_2 = 2$	[6]
CO4/SO1/BL3	c.	Determine whether Eulerian circuit, Hamiltonian path and Hamiltonian cycle exists for the following graph. If so, find them.  	[8]